

Total No. of Questions: 22

Total No. of Pages: 02

Roll No. _____

Paper Code
3E1203

B.Tech. III-Sem (Main & Back) 2025-26
Artificial Intelligence
3CS3-04 Digital Electronics
AID,CAI,CS,IT,CCS, CDS,CIT,CSD,CSR,AI
3E1203

Time:3 Hours

Maximum Marks :70

Attempt all ten questions from Part A. Attempt any Five questions out of seven questions from Part B and three questions out of five questions from Part C.

Schematic diagrams must be shown wherever necessary. Any data missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting material is permitted during examination. (As mentioned in form No. 205)

1- _____

2- _____

Part A

(Answer should be given up to 25 words only)

All questions are compulsory.

10x2=20

- Q.1 Convert the binary number $(10110,11)_2$ to its decimal equivalent.
- Q.2 What is the 1's complement of the binary number 11010100_2 ?
- Q.3 State the De Morgan's Theorem for two variables A and B.
- Q.4 Define a minterm for a Boolean expression.
- Q.5 What is the purpose of a "Don't Care" condition in a Karnaugh Map?
- Q.6 List two characteristics of TTL logic gates.
- Q.7 What is the main function of an encoder?
- Q.8 Differentiate between a latch and a flip-flop.
- Q.9 How many flip-flops are required to construct a MOD-12 counter.
- Q.10 What is a shift register?

Part B

(Analytical/Problem Solving Questions)

Attempt any Five questions

5x4=20

- Q.1 Discuss the types of Registers with suitable examples and diagrams.
- Q.2 Simplify the following Boolean expression using Boolean algebra theorems.

$$F(A,B,C) = A'B'C' + A'BC' + AB'C' + ABC'$$
- Q.3 Draw the circuit diagram of a 2-input TTL NAND gate and briefly explain its operation.
- Q.4 Design a half-adder circuit using basic logic gates and write its truth table.
- Q.5 Explain the operation of a D flip-flop with its characteristic table.
- Q.6 Convert the following SOP expression to POS form: $F(A,B,C) = \sum m(0,1,3,7)$
- Q.7 Explain the concept of "Fan-out" in digital logic families.

Part C

(Descriptive/Analytical/Problem solving/Design questions)

Attempt any Three questions .

3x10=30

- Q.1 (a) Explain the working principle of a 4-to-1 Multiplexer. Draw its logic diagram.
- (b) Implement the following Boolean function using a 4-to-1 Multiplexer:

$$F(A,B,C) = \sum m(1,2,5,7)$$
- Q.2 Discuss the counters with the state diagram, stat table, and the final logic circuit.
- Q.3 Explain the Quine-McCluskey method for Boolean function minimization. What are its advantages over the K-map method for more variables.
- Q.4 Minimize the following Boolean function using the Karnaugh Map (K-Map) method and draw the logic circuit using NAND gates only:

$$F(A,B,C,D) = \sum m(0,2,5,7,8,10,13,15)$$
- Q.5 Differentiate between RTL, DTL, and CMOS logic families based on their characteristics such as power dissipation, propagation delay, and noise margin.